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REFRIGERATOR WITH MACHINE ROOM OF WHICH COVER HAS ATTENUATED VIBRATION

Abstract

A refrigerator includes a machine room provided at a lower portion of a main body; an opening provided on one side of the main body to allow access to the machine room; a cover disposed on one side of the main body to cover the opening; a plurality of cover catches provided on a lower portion of the cover; an emboss provided between the plurality of cover catches on the lower portion of the cover; and a plurality of main body hooks provided below the opening on one side of the main body such that locations of the plurality of main body hooks and the plurality of cover catches of the cover correspond to each other.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation application is a continuation application, under 35 U.S.C. § 111(a), of international application No. PCT/KR2023/018583, filed Nov. 17, 2023, which claims priority under 35 U. S. C. § 119 to Korean Patent Application No. 10-2022-0174222, filed Dec. 13, 2023, the disclosures of which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

[0002] The disclosure relates to a refrigerator, and more particularly, to a refrigerator capable of damping vibration of a cover covering a machine room.

BACKGROUND ART

[0003] A refrigerator is a device that cools food or stores it at low temperature to prevent spoilage or deterioration of food.

[0004] The refrigerator may include a storage space that accommodates food and a cooling device that cools the storage space.

[0005] Generally, a refrigeration cycle device that generates cold air using a refrigeration cycle is used as the cooling device.

[0006] The refrigeration cycle device may generate cold air by circulating a refrigerant through a closed circuit including a compressor, a condenser, an expansion device, and an evaporator.

[0007] Therefore, the refrigerator may include a machine room in which the compressor and the condenser are accommodated. An opening communicating with the machine room may be provided on one side of the refrigerator. A cover may be detachably disposed in the opening of the machine room.

DISCLOSURE OF INVENTION

Technical Problem

[0008] The purpose of the disclosure is to provide a refrigerator capable of preventing or minimizing vibration noise of a cover caused by vibration of a compressor disposed in a machine room.

[0009] In addition, the purpose of the disclosure is to provide a refrigerator capable of reducing the vibration-proof cost for preventing vibration noise of a cover.

Technical Solution

[0010] According to an embodiment of the disclosure, a refrigerator may include: a machine room to be provided at a lower portion of a main body; an opening to be provided on one side of the main body to allow access to the machine room; a cover to be disposed on one side of the main body to cover the opening; a plurality of cover catches to be provided on a lower portion of the cover; an emboss to be provided between the plurality of cover catches on the lower portion of the cover; and a plurality of main body hooks to be provided on the one side of the main body below the opening such that while the plurality of main body hooks are provided on the side of the main body, locations of the plurality of main body hooks and the plurality of cover catches of the cover.

[0011] While the opening of the machine room being covered with the cover, the plurality of cover catches may be caught on the plurality of main body hooks, and the emboss may contact the one side of the main body on which the plurality of main body hooks are formed.

[0012] A height of the emboss may be greater than a gap between the cover and the one side of the main body.

[0013] Each of the plurality of main body hooks may be formed to be inclined upwardly on one side of the main body.

[0014] The emboss may include a lower inclined surface formed to be inclined upwardly and facing a lower end of the cover.

[0015] The emboss may be formed to have an arc cross-section or a triangular cross-section.

[0016] The emboss may be among a plurality of embosses arranged in a straight line.

[0017] The cover may be formed as a rectangular flat plate. The plurality of cover catches may be provided on a lower portion of the cover. Each of the plurality of cover catches may be formed in a protrusion shape protruding from one side of the cover. A catch groove into which the main body hook is insertable may be provided at a lower end of the cover catch.

[0018] The emboss may be formed to protrude from one side of the cover in a same direction as the plurality of cover catches.

[0019] A height of the emboss may be higher than a height of the plurality of cover catches.

[0020] The refrigerator may further include: a fastening member formed to fasten an upper portion of the cover to one side of the main body above the opening.

[0021] The main body may include an upper support part, a lower support part, a left support part, and a right support part provided at an edge of the opening. The plurality of main body hooks may be provided on the lower support part. A plurality of screw holes may be provided on the upper support part.

[0022] The cover may include an upper fixing portion and a lower fixing portion. The plurality of cover catches may be provided on the lower fixing portion. A plurality of bolt holes corresponding to the plurality of screw holes of the upper support part of the main body **10** may be provided on the upper fixing portion.

[0023] According to an embodiment of the disclosure, a refrigerator may include: a machine room to be provided at a lower portion of a main body; an opening to be provided on one side of the main body to allow access to the machine room; a cover to be disposed on one side of the main body to cover the opening; a plurality of main body hooks to be provided on one side of the main body below the opening; an emboss to be provided on one side of the main body between the plurality of main body hooks and below the opening; and a plurality of cover catches to be provided on a lower portion of the cover such that while the cover is disposed on the one side of the main body, locations of the cover catches and the plurality of main body hooks of the main body correspond to each other.

[0024] Each of the plurality of cover catches may be formed to be inclined downward on a lower portion of the cover. Each of the plurality of main body hooks may be formed in a protrusion shape protruding from one side of the main body. A catch groove into which the cover catch is insertable may be provided on an upper surface of the main body hook.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0025] These and/or other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0026] FIG. **1** is a rear perspective view illustrating a refrigerator **1** according to an embodiment of the disclosure.

[0027] FIG. **2** is a partial enlarged perspective view illustrating the refrigerator **1** of FIG. **1** according to an embodiment of the disclosure.

[0028] FIG. 3 is a perspective view illustrating a state in which a cover 40 is separated from the refrigerator 1 of FIG. 2 according to an embodiment of the disclosure.

[0029] FIG. 4 is a partial perspective view illustrating a machine room of a refrigerator according to an embodiment of the disclosure.

[0030] FIG. 5 is an enlarged perspective view of part A of FIG. 4 according to an embodiment of the disclosure.

[0031] FIG. 6 is a perspective view illustrating a cover configured to cover an opening of a machine room of a refrigerator according to an embodiment of the disclosure.

[0032] FIG. 7 is a perspective view illustrating the cover of FIG. 6 as viewed from the rear according to an embodiment of the disclosure.

[0033] FIG. 8 is an enlarged perspective view of part B of FIG. 7 according to an embodiment of the disclosure.

[0034] FIG. 9 is an enlarged perspective view of part C of FIG. 2 according to an embodiment of the disclosure.

[0035] FIG. 10 is a cross-sectional view taken along line C-C of FIG. 9 according to an embodiment of the disclosure.

[0036] FIG. 11 is a cross-sectional view taken along line D-D of FIG. 9 according to an embodiment of the disclosure.

[0037] FIG. 12A is a partial cross-sectional view illustrating a state before assembling a cover to a machine room of a refrigerator according to an embodiment of the disclosure.

[0038] FIG. 12B is a partial cross-sectional view illustrating a state in which a cover catch of a cover is coupled to a main body hook of a machine room of a refrigerator according to an embodiment of the disclosure.

[0039] FIG. 12C is a partial cross-sectional view illustrating a state in which a cover is assembled to a machine room of a refrigerator according to an embodiment of the disclosure.

[0040] FIG. 13 is a view illustrating forces acting between a machine room and a cover when the cover is disposed in the machine room of a refrigerator according to an embodiment of the disclosure.

[0041] FIG. 14 is a perspective view illustrating another example of a machine room of a refrigerator according to an embodiment of the disclosure.

[0042] FIG. 15 is a perspective view illustrating a cover disposed in the machine room of the refrigerator of FIG. 14 according to an embodiment of the disclosure.

BEST MODE FOR CARRYING OUT THE INVENTION

[0043] Since the embodiments of the disclosure can apply various transformations and have various embodiments, specific embodiments will be illustrated in the drawings and described in detail in the detailed description. However, this is not intended to limit the scope to the specific embodiments, and should be understood to include various modifications, equivalents, and/or alternatives of the embodiment of the disclosure. In connection with the description of the drawings, like reference numerals may be used for like elements.

[0044] In describing the disclosure, when it is determined that a detailed description of a related known function or configuration may unnecessarily obscure the gist of the disclosure, a detailed description thereof will be omitted.

[0045] In addition, the following embodiments may be modified in many different forms, and the scope of the technical idea of the disclosure is not limited to the following embodiments. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the spirit of the disclosure to those skilled in the art.

[0046] Terms used in this disclosure are only used to describe specific embodiments, and are not intended to limit the scope of rights. Singular expressions include plural expressions unless the context clearly dictates otherwise.

[0047] In this disclosure, expressions such as “has,” “can have,” “includes,” or “can include”

indicate the existence of a corresponding feature (e.g., numerical value, function, operation, or component such as a part) and do not preclude the existence of additional features.

[0048] In this disclosure, expressions such as “A or B,” “at least one of A or/and B,” or “one or more of A or/and B” may include all possible combinations of the items listed together. For example, “A or B,” “at least one of A or/and B,” or “one or more of A or/and B” may refer to all cases (1) including at least one A, (2) including at least one B, or (3) including both at least one A and at least one B.

[0049] Expressions such as “first,” “second,” “primary,” or “secondary,” as used in this disclosure may modify various components regardless of order and/or importance, are used only to distinguish one component from other components, and do not limit the corresponding components.

[0050] Further, terms such as ‘leading end’, ‘rear end’, ‘upper side’, ‘lower side’, ‘top end’, ‘bottom end’, etc. used in the disclosure are defined with reference to the drawings. However, the shape and position of each component are not limited by these terms.

[0051] Hereinafter, embodiments of a refrigerator **1** according to the disclosure will be described in detail with reference to the attached drawings.

[0052] FIG. **1** is a rear perspective view illustrating a refrigerator **1** according to an embodiment of the disclosure.

[0053] Referring to FIG. **1**, a refrigerator **1** according to an embodiment of the disclosure may include a main body **10**.

[0054] The main body **10** may form the exterior of the refrigerator **1**. A refrigerating compartment and a freezing compartment for accommodating food may be provided inside the main body **10**.

[0055] A refrigerating compartment opening communicating with the refrigerating compartment and a freezing compartment opening communicating with the freezing compartment may be provided on the front side of the main body **10**. In addition, a refrigerating compartment door and a freezing compartment door for opening and closing the refrigerating compartment opening and the freezing compartment opening may be provided on the front side of the main body **10**.

[0056] The main body **10** may be provided with a refrigeration cycle device capable of cooling the refrigerating compartment and the freezing compartment.

[0057] The refrigeration cycle device may include a compressor, a condenser, an expansion valve, and a heat exchanger.

[0058] The heat exchanger may be disposed adjacent to the refrigerating compartment and the freezing compartment.

[0059] The compressor, the condenser, and the expansion valve may be disposed in a machine room **20** provided at a lower portion of the main body **10**. In other words, the machine room **20** may be provided at the lower portion of the main body **10**, and the compressor, the condenser, and the expansion valve may be disposed in the machine room **20**.

[0060] The machine room **20** may be covered by a cover **40** that is detachable. Therefore, when the cover **40** is opened, the machine room **20** is exposed, thereby allowing the worker to access the machine room **20**.

[0061] Hereinafter, the machine room **20** of the refrigerator **1** according to an embodiment of the disclosure will be described in detail with reference to FIGS. **2** to **5**.

[0062] FIG. **2** is a partial enlarged perspective view illustrating the refrigerator **1** of FIG. **1**. FIG. **3** is a perspective view illustrating a state in which a cover **40** is separated from the refrigerator **1** of FIG. **2**. FIG. **4** is a partial perspective view illustrating a machine room **20** of a refrigerator **1** according to an embodiment of the disclosure. FIG. **5** is an enlarged perspective view of part A of FIG. **4**.

[0063] Referring to FIGS. **2** to **4**, the machine room **20** may be provided at the lower portion of the main body **10** of the refrigerator **1**. In other words, the machine room **20** may be provided adjacent to the bottom of the refrigerator **1**, that is, the bottom of the main body **10**. The machine room **20**

may be formed of a chassis and have a space having a roughly rectangular parallelepiped shape.
[0064] The compressor **11**, the condenser, and the expansion valve may be disposed in the machine room **20**. In the machine room **20** of FIGS. **3** and **4**, only the compressor **11** that generates vibration is shown for the convenience of illustration.

[0065] An opening **21** communicating with the machine room **20** may be provided on one side of the main body **10**. In the case of this embodiment, the opening **21** may be provided on the rear side of the main body **10**. The worker may access the compressor **11**, the condenser, and the expansion valve disposed in the machine room **20** through the opening **21**.

[0066] The opening **21** may be formed in a shape corresponding to the machine room **20**. For example, the opening **21** of the machine room **20** may be formed in an approximately rectangular shape.

[0067] The opening **21** of the machine room **20** may be covered with the cover **40**. In other words, the cover **40** may be disposed on one side of the main body **10** so as to cover the opening **21** of the machine room **20**.

[0068] The machine room **20** may include a support part **22** supporting the cover **40**. The support part **22** may be provided to contact the cover **40** at the edge of the opening **21** of the machine room **20**. The support part **22** may be formed along the entire circumference of the opening **21**. The outer surface of the support part **22** in contact with the cover **40** may be formed in a plane.

[0069] The support part **22** may be formed as a part of the chassis forming the machine room **20**. For example, the part of the chassis forming the opening **21** of the machine room **20** may form the support part **22**.

[0070] The support part **22** of the machine room **20** may include an upper support part **221**, a lower support part **222**, a left support part **223**, and a right support part **224**.

[0071] The upper support part **221** may be provided on the upper side of the opening **21**. A plurality of screw holes **23** may be provided on the upper support part **221**. In other words, the plurality of screw holes **23** may be provided at the upper side of the opening **21** on one side of the main body **10**. A female screw may be provided on the inner surface of each of the plurality of screw holes **23**.

[0072] The left support part **223** may be provided on the left side of the opening **21**. The right support part **224** may be provided on the right side of the opening **21**.

[0073] The lower support part **222** may be provided on the lower side of the opening **21**. A plurality of main body hooks **30** may be provided on the lower support part **222**. In other words, the plurality of main body hooks **30** may be provided below the opening **21** on one side of the main body **10**. The plurality of main body hooks **30** may be formed to support the cover **40**.

[0074] The plurality of main body hooks **30** may be disposed at a certain interval on the lower support part **222**. Each of the plurality of main body hooks **30** may be formed to be inclined upward on one side of the main body **10**. The plurality of main body hooks **30** may be formed below the lower end of the opening **21**.

[0075] The plurality of main body hooks **30** may be formed in the same shape, so one main body hook **30** will be described below.

[0076] Referring to FIG. **5**, the main body hook **30** may include a hook portion **31** and a guide portion **32**.

[0077] The hook portion **31** may protrude outward from one side of the main body **10**. The hook portion **31** may be formed approximately perpendicular to one side of the main body **10**. The hook portion **31** may be formed approximately in a rectangular shape. The hook portion **31** may be formed with the same thickness as the chassis forming the main body **10**.

[0078] The guide portion **32** may extend upward from the tip of the hook portion **31**. The guide portion **32** may be formed so that the upper end of the guide portion **32** is positioned lower than the lower end of the opening **21**.

[0079] The guide portion **32** may be formed so that the width thereof becomes narrower from the bottom to the top. In other words, the width of the lower end of the guide portion **32** connected to

the hook portion **31** may be larger than the width of the upper end of the guide portion **32**. For example, the guide portion **32** may be formed in an approximately triangular shape.

[0080] The guide portion **32** may be formed so as to be inclined upward from the tip of the hook portion **31**. When the guide portion **32** is formed so as to be inclined upward, the cover **40** may be easily coupled to the main body hook **30**.

[0081] The main body hook **30** may be formed integrally with the main body **10**. For example, the main body hook **30** may be formed by press-processing the lower support part **222** of the machine room **20** formed on the main body **10**.

[0082] The cover **40** may be coupled to the plurality of main body hooks **30**. The cover **40** may be formed to cover the opening **21** of the machine room **20** of the main body **10**.

[0083] Hereinafter, the cover **40** will be described in detail with reference to FIGS. **6** to **8**.

[0084] FIG. **6** is a perspective view illustrating a cover **40** configured to cover an opening **21** of a machine room **20** of a refrigerator **1** according to an embodiment of the disclosure. FIG. **7** is a perspective view illustrating the cover **40** of FIG. **6** as viewed from the rear. FIG. **8** is an enlarged perspective view of part B of FIG. **7**.

[0085] Referring to FIGS. **6** and **7**, the cover **40** may be formed as a flat plate having a thin thickness and a roughly rectangular shape. The cover **40** may be formed of an elastic material. For example, the cover **40** may be formed of a thin metal plate.

[0086] The cover **40** may include a cover portion **41** and a fixing portion **42**.

[0087] The cover portion **41** may be formed in a roughly rectangular shape so as to cover the opening **21** of the machine room **20**. The cover portion **41** may include a plurality of air holes **45**. The plurality of air holes **45** may be formed as slits. External air may be introduced into the machine room **20** through the plurality of air holes **45**.

[0088] The fixing portion **42** may be provided at the edge of the cover portion **41**. The fixing portion **42** may be formed along the entire circumference of the cover portion **41**. The fixing portion **42** may be formed to correspond to the support part **22** of the machine room **20**.

[0089] In order to increase the rigidity of the cover **40**, the fixing portion **42** and the cover portion **41** may be formed to have a step. For example, the cover portion **41** may be formed as an approximately rectangular groove protruding from the fixing portion **42**.

[0090] The fixing portion **42** may include an upper fixing portion **421**, a lower fixing portion **422**, a left fixing portion **423**, and a right fixing portion **424**.

[0091] The upper fixing portion **421** may be provided on the upper portion of the cover **40**, that is, on the upper side of the cover portion **41**. The upper fixing portion **421** may be formed to correspond to the upper support part **221** of the machine room **20**. The upper fixing portion **421** may contact the upper support part **221** of the machine room **20**.

[0092] The upper fixing portion **421** may include a plurality of bolt holes **43** corresponding to the plurality of screw holes **23** of the upper support part **221**. Accordingly, the upper fixing portion **421** of the cover **40** may be fixed to the upper support part **221** of the main body **10** with a plurality of bolts.

[0093] The left fixing portion **423** may be provided on the left side of the cover portion **41**. The right fixing portion **424** may be provided on the right side of the cover portion **41**.

[0094] The lower fixing portion **422** may be provided on the lower portion of the cover **40**, that is, on the lower side of the cover portion **41**. The lower fixing portion **422** may be formed to correspond to the lower support part **222** of the machine room **20**. The lower fixing portion **422** may contact the lower support part **222** of the machine room **20**.

[0095] The lower fixing portion **422** may include a plurality of cover catches **50**. In other words, the plurality of cover catches **50** may be provided at the lower portion of the cover **40**. The plurality of cover catches **50** may be provided below the cover portion **41**.

[0096] The plurality of cover catches **50** may be provided to correspond to the plurality of main body hooks **30** provided on one side of the main body **10**. In detail, the plurality of cover catches

50 may be formed so as to be caught on the plurality of main body hooks **30** provided on the lower support part **222** of the machine room **20**.

[0097] The plurality of cover catches **50** may be formed in the same shape, so one cover catch **50** will be described below.

[0098] Referring to FIG. **8**, the cover catch **50** may be formed in a protrusion shape protruding from one side of the cover **40**. In other words, the cover catch **50** may be formed in a protrusion shape protruding from the lower fixing portion **422** of the cover **40** toward the machine room **20**. Accordingly, the cover catch **50** may protrude from one surface of the fixing portion **42** in a direction opposite to the cover portion **41**.

[0099] A catch groove **53** may be provided at the lower end **51** of the cover catch **50**. The catch groove **53** of the cover catch **50** may be formed so that the main body hook **30** is inserted into the catch groove **53**. In other words, the guide portion **32** of the main body hook **30** may be inserted into the catch groove **53** of the cover catch **50**.

[0100] In the case of the embodiment shown in FIG. **8**, the cover catch **50** may be formed in a U-shape with a substantially flat bottom. In other words, the cover catch **50** may be formed in a channel shape. The upper end **52** and the lower end **51** of the cover catch **50** may be open. The open lower end **51** of the cover catch **50** may form the catch groove **53**. Therefore, when the cover **40** is disposed in the opening **21** of the machine room **20**, the guide portion **32** of the main body hook **30** may be inserted into the catch groove **53** of the cover catch **50**.

[0101] The inner width **W2** of the cover catch **50** may be formed in a size corresponding to the width **W1** of the main body hook **30** (see FIG. **5**). In other words, the inner width **W2** of the cover catch **50** may be formed to be larger than the width **W1** of the main body hook **30**.

[0102] The left-right movement amount of the cover **40** may be defined by the cover catches **50** and the main body hooks **30**. In other words, the left-right movement amount of the cover **40** may be defined by the difference between the inner width **W2** of the cover catch **50** and the width **W1** of the main body hook **30**.

[0103] FIG. **9** is an enlarged perspective view of part C of FIG. **2**. FIG. **10** is a cross-sectional view taken along line C-C of FIG. **9**.

[0104] Referring to FIGS. **9** and **10**, the cover catch **50** may be coupled to the main body hook **30**. In detail, the guide portion **32** of the main body hook **30** may be inserted into the catch groove **53** of the cover catch **50**, and the lower end **51** of the cover catch **50** may be supported by the hook portion **31** of the main body hook **30**. In addition, the cover catch **50** may be located between the guide portion **32** of the main body hook **30** and the lower support part **222** of the machine room **20**.

[0105] Accordingly, the left-right movement of the cover **40** may be limited by the guide portion **32** of the main body hook **30**, and the downward movement of the cover **40** may be limited by the hook portion **31** of the main body hook **30**. In addition, the forward-backward movement of the cover **40**, that is, the movement of the cover **40** in the direction perpendicular to the rear side of the refrigerator **1**, may be limited by the guide portion **32** of the main body hook **30**.

[0106] When the cover **40** is assembled to the machine room **20**, that is, when the cover catches **50** of the cover **40** are coupled to the main body hooks **30** of the machine room **20**, there may be a gap **G** between the lower fixing portion **422** of the cover **40** and the lower support part **222** of the machine room **20**. The gap **G** between the cover **40** and the support part **22** of the machine room **20** may be defined by the cover catches **50** and the main body hooks **30**. For example, the gap **G** between the cover **40** and the support part **22** of the machine room **20** may be defined to be the same as the height **H1** of the cover catch **50**.

[0107] The cover **40** may include an emboss **60**. The emboss **60** may be provided on the lower portion of the cover **40**, that is, the lower fixing portion **422**. The emboss **60** may be provided below the cover portion **41**.

[0108] The emboss **60** may be formed to protrude from one surface of the cover **40** in the same direction as the plurality of cover catches **50**.

[0109] The emboss **60** may be provided between the plurality of cover catches **50**. The emboss **60** may be provided between two adjacent cover catches **50** among the plurality of cover catches **50**. [0110] As another example, the emboss **60** may include a plurality of embosses **60**. For example, when the cover **40** includes four cover catches **50**, three embosses **60** may be formed between the four cover catches **50**. When the plurality of embosses **60** are formed, the plurality of embosses **60** may be formed to form a straight line.

[0111] The emboss **60** may include a lower inclined surface **61** that faces the lower end of the cover **40** and is formed to be inclined upward. In detail, one surface of the emboss **60** that faces the lower end of the cover **40** may be formed to be inclined upward with respect to the cover **40**. In other words, the emboss **60** may include the lower inclined surface **61** that is inclined upward with respect to the cover **40**.

[0112] For this purpose, the emboss **60** may be formed to have an arc cross-section or a triangular cross-section. In other words, the emboss **60** may be formed as a column having an arc cross-section or a triangular column. However, the shape of the emboss **60** is not limited thereto. The emboss **60** may be formed as a column having various cross-sections having the lower inclined surface **61**.

[0113] As described above, in the case where the emboss **60** includes the lower inclined surface **61**, when assembling the cover **40** to the machine room **20**, the emboss **60** may be prevented from interfering with the lower support part **222** of the machine room **20**.

[0114] The emboss **60** may be formed higher than the height **H1** of the plurality of cover catches **50**. In other words, the height **H2** of the emboss **60** may be higher than the height **H1** of the cover catches **50**. When the height **H2** of the emboss **60** is made higher than the height **H1** of the cover catches **50**, the tip of the emboss **60** may be in close contact with the support part **22** of the machine room **20**. Accordingly, it is possible to prevent a gap from being formed between the support part **22** of the machine room **20** and the fixing portion **42** of the cover **40**.

[0115] FIG. **11** is a cross-sectional view taken along line D-D of FIG. **9**.

[0116] Referring to FIG. **11**, when the cover catch **50** of the cover **40** is caught on the main body hook **30** of the machine room **20**, the tip of the emboss **60** may be in contact with the lower support part **222** of the machine room **20**. At this time, because the height **H2** of the emboss **60** is greater than the gap **G** between the cover **40** and the support part **22** of the machine room **20**, that is, the lower fixing portion **422** of the cover **40** and the lower support part **222** of the machine room **20**, the emboss **60** may be in close contact with the support part **22** of the machine room **20**.

[0117] In other words, the cover **40** may be pressed against the support part **22** of the machine room **20** with a force applied to it by the difference between the gap **G** between the cover **40** and the support part **22** of the machine room **20** and the height **H2** of the emboss **60**. Therefore, there is no gap between the cover **40** and the support part **22** of the machine room **20**, that is, between the tip of the emboss **60** of the cover **40** and the lower support part **222** of the machine room **20**.

[0118] The cover catches **50** and the emboss **60** may be formed integrally with the cover **40**. For example, when manufacturing the cover **40**, the lower fixing portion **422** may be press-processed to form the plurality of cover catches **50** and the emboss **60**.

[0119] The lower portion of the cover **40** may be fixed to the lower support part **222** of the machine room **20** by the cover catches **50** described above, and the upper portion of the cover **40** may be fixed to the upper support part **221** of the machine room **20** by the fastening members.

[0120] In the case of this embodiment, a plurality of bolts may be used as the fastening members. In other words, by inserting the plurality of bolts into the plurality of bolt holes **43** of the cover **40** and fastening them to the plurality of screw holes **23** provided in the support part **22** of the machine room **20**, the upper portion of the cover **40** may be fixed to the main body **10** above the opening **21** of the machine room **20**.

[0121] However, the fastening member that secures the upper portion of the cover **40** to the upper support part **221** of the machine room **20** is not limited thereto. Any type of fastening member may

be used as long as the upper portion of the cover **40** can be secured to the upper support part **221** of the machine room **20**. For example, a one-touch coupling structure may be used as the fastening member.

[0122] Hereinafter, a method of assembling the cover **40** to the machine room **20** of the main body **10** according to an embodiment of the disclosure will be described with reference to FIGS. **12A** to **12C**.

[0123] FIG. **12A** is a partial cross-sectional view illustrating a state before assembling a cover **40** to a machine room **20** of a refrigerator **1** according to an embodiment of the disclosure. FIG. **12B** is a partial cross-sectional view illustrating a state in which a cover catch **50** of a cover **40** is coupled to a main body hook **30** of a machine room **20** of a refrigerator **1** according to an embodiment of the disclosure. FIG. **12C** is a partial cross-sectional view illustrating a state in which a cover **40** is assembled to a machine room **20** of a refrigerator **1** according to an embodiment of the disclosure.

[0124] Referring to FIG. **12A**, the cover **40** may be positioned obliquely in front of the machine room **20** of the main body **10**. At this time, the cover **40** may be positioned so that the plurality of cover catches **50** face downward. In addition, the cover **40** may be positioned so that the plurality of cover catches **50** of the cover **40** correspond to the plurality of main body hooks **30** of the main body **10**.

[0125] In the state of FIG. **12A**, the cover **40** may be moved downward in the direction of the arrow. In other words, the main body hooks **30** of the main body **10** may be obliquely inserted into the cover catches **50** of the cover **40**. Then, as illustrated in FIG. **12B**, the plurality of cover catches **50** of the cover **40** may be caught on the plurality of main body hooks **30** of the main body **10**. In detail, the lower ends of the plurality of cover catches **50** of the cover **40** may be positioned on the hook portions **31** of the main body hooks **30** of the main body **10**. In other words, the plurality of cover catches **50** may be positioned between the lower support part **222** of the main body **10** and the guide portions **32** of the main body hooks **30**.

[0126] In other words, the upper and lower positions and the left and right positions of the cover **40** may be defined by the plurality of main body hooks **30** of the machine room **20** and the plurality of cover catches **50** of the cover **40**.

[0127] In the state of FIG. **12B**, the upper portion of the cover **40** may be moved to the left in the direction of the arrow. In other words, the upper portion of the cover **40** may be pushed toward the machine room **20**. Then, as illustrated in FIG. **12C**, the upper fixing portion **421** of the cover **40** may come into contact with the upper support part **221** of the machine room **20**.

[0128] In this state, when the cover **40** is fixed to the main body **10** with the plurality of fastening members, the assembly of the cover to the front side of the machine room **20** of the main body **10** may be completed. For example, the upper fixing portion **421** of the cover **40** may be fixed to the upper support part **221** of the main body **10** using the plurality of bolts.

[0129] As described above, when the cover **40** is assembled to the support part **22** of the main body **10**, the cover **40** is in close contact with the support part **22** of the main body **10**, so that there may be no gap between the cover **40** and the support part **22** of the main body **10**. Hereinafter, the force acting between the lower fixing portion **422** of the cover **40** and the lower support part **222** of the main body **10** in the state where the cover **40** is assembled to the main body **10** will be described in detail with reference to FIG. **13**.

[0130] FIG. **13** is a view illustrating forces acting between a machine room **20** and a cover **40** when the cover **40** is disposed in the machine room **20** of a refrigerator **1** according to an embodiment of the disclosure.

[0131] Referring to FIG. **13**, the lower end **51** of the cover catch **50** may be caught on the main body hook **30**, that is, a portion connecting the hook portion **31** and the guide portion **32**, so that the cover catch **50**, that is, the cover **40**, may be applied with force in the direction of arrow **F1**.

[0132] In addition, because the tip of the emboss **60** of the cover **40** is in contact with the lower support part **222** of the main body **10**, force may be applied to the emboss **60**, that is, the cover **40**,

in the direction of arrow F2. In detail, because the height H2 of the emboss 60 is greater than the gap G between the lower fixing portion 422 of the cover 40 and the lower support part 222 of the main body 10 or the height H1 of the cover catch 50, force may be applied to the emboss 60 of the cover 40 in the direction of arrow F2.

[0133] Accordingly, the cover 40 may be forcibly fitted to the main body hooks 30 of the main body 10 by the force in the F1 direction applied to the cover catch 50 and the force in the F2 direction applied to the emboss 60. Therefore, no gap may occur between the emboss 60 of the cover 40 and the lower support part 222 of the main body 10.

[0134] The compressor 11 (see FIG. 4) may be disposed in the machine room 20. Accordingly, when the compressor 11 operates, vibration may occur. This vibration may be transmitted to the lower support part 222 through the chassis forming the machine room 20 in which the compressor 11 is disposed.

[0135] The opening 21 of the machine room 20 may be covered with the cover 40. Therefore, the lower support part 222 of the machine room 20 may face the lower fixing portion 422 of the cover 40. At this time, when there is a gap between the lower support part 222 of the main body 10 and the cover 40, noise may be generated as the cover 40 vibrates slightly due to the vibration of the compressor 11.

[0136] However, in the refrigerator 1 according to an embodiment of the disclosure, because the lower support part 222 of the machine room 20 and the lower fixing portion 422 of the cover 40 are forcibly fitted by the emboss 60 and the cover catches 50, noise in which the cover 40 vibrates slightly due to the vibration of the compressor 11, that is, vibration noise may not occur.

[0137] In general, in order to prevent the vibration noise of the cover 40, a vibration-proof member may be disposed between the cover 40 and the main body 10. However, because the vibration-proof member is formed of a material such as sponge, rubber, and the like, there may be a problem that the gap between the cover 40 and the main body 10 may not be maintained uniformly. When the gap between the cover 40 and the main body 10 is not uniform, the vibration noise of the cover 40 may occur.

[0138] In addition, when the vibration-proof member is used, there may be a problem that the material cost of the vibration-proof member and the cost of attaching the vibration-proof member to the cover 40 are incurred.

[0139] Alternatively, the cover 40 may be coupled to the main body 10 using a plurality of bolts to prevent the vibration noise of the cover 40. However, when the threads of the bolts are broken during assembly, a gap may occur between the cover 40 and the main body 10. In addition, when the positions of the screw holes of the main body 10 and the positions of the bolt holes of the cover 40 do not match, a gap may occur between the cover 40 and the main body 10. When the gap occurs between the cover 40 and the main body 10, the vibration noise of the cover 40 may occur.

[0140] In addition, when a number of bolts are used, there may be a problem that the material cost and assembly cost are incurred as much as the number of bolts.

[0141] However, in the case of the disclosure, the cover 40 may be tightly attached to the main body 10 without a gap by using the plurality of main body hooks 30 provided on the main body 10 and the plurality of cover catches 50 and emboss 60 provided on the cover 40. Therefore, the refrigerator 1 according to an embodiment of the disclosure may reduce the material cost and assembly cost of the cover 40 compared to the refrigerator according to the prior art.

[0142] In the above, the case where the emboss 60 is provided on the lower fixing portion 422 of the cover 40 has been described. However, the disclosure is not limited thereto. Conversely, the emboss 60 may be provided on the lower support part 222 of the main body 10. Hereinafter, a refrigerator 1 in which the emboss 60 is provided on the main body 10 will be described with reference to FIGS. 14 and 15.

[0143] FIG. 14 is a perspective view illustrating another example of a machine room 20 of a refrigerator 1 according to an embodiment of the disclosure. FIG. 15 is a perspective view

illustrating a cover **40** disposed in the machine room **20** of the refrigerator **1** of FIG. **14**.

[0144] Referring to FIG. **14**, the lower support part **222** may be provided at the lower side of the opening **21** of the machine room **20**. The plurality of main body hooks **30** may be provided on the lower support part **222**. In other words, the plurality of main body hooks **30** may be provided below the opening **21** on one side of the main body **10**. The plurality of main body hooks **30** may be formed to support the cover **40**.

[0145] The plurality of main body hooks **30** may be disposed at a certain interval on the lower support part **222**. The plurality of main body hooks **30** may be formed in the same shape.

[0146] The main body hook **30** may be formed in a protrusion shape protruding from one side of the main body **10**. In other words, the main body hook **30** may be formed in a protrusion shape protruding outward from the lower support part **222** of the main body **10**.

[0147] A catch groove **33** may be provided at the upper end of the main body hook **30**. The catch groove **33** of the main body hook **30** may be formed so that a cover catch **50** is inserted into the catch groove **33**.

[0148] In the case of the embodiment shown in FIG. **14**, the main body hook **30** may be formed in a U-shape with a substantially flat bottom. In other words, the main body hook **30** may be formed in a channel shape. The upper end **34** and lower end **35** of the main body hook **30** may be open. The open upper end **34** of the main body hook **30** may form the catch groove **33**. Therefore, when disposing the cover **40** in the opening **21** of the machine room **20**, the cover catch **50** may be inserted into the catch groove **33** of the main body hook **30**.

[0149] An emboss **60** may be provided on the lower support part **222** of the main body **10**. In other words, the emboss **60** may be provided below the opening **21** of the machine room **20**.

[0150] The emboss **60** may be formed to protrude from one side of the main body **10** in the same direction as the plurality of main body hooks **30**.

[0151] The emboss **60** may be provided between the plurality of main body hooks **30**. The emboss **60** may be provided between two adjacent main body hooks **30** among the plurality of main body hooks **30**.

[0152] As another example, the emboss **60** may include a plurality of embosses **60**. For example, when four main body hooks **30** are provided on the main body **10**, three embosses **60** may be formed between the four main body hooks **30**. When the plurality of embosses **60** are provided, the plurality of embosses **60** may be formed to form a straight line.

[0153] The emboss **60** may include an upper inclined surface **62** that faces the lower end of the opening **21** and is formed to be inclined downward. In detail, one surface of the emboss **60** that faces the lower end of the opening **21** may be formed to be inclined downward with respect to the lower support part **222** of the main body **10**. In other words, the emboss **60** may include the upper inclined surface **62** that is inclined downward with respect to the lower support part **222** of the main body **10**.

[0154] For this purpose, the emboss **60** may be formed to have an arc cross-section or a triangular cross-section. In other words, the emboss **60** may be formed as a column having an arc cross-section or a triangular column. However, the shape of the emboss **60** is not limited thereto. The emboss **60** may be formed as a column having various cross-sections having the upper inclined surface **62**.

[0155] As described above, in the case where the emboss **60** includes the upper inclined surface **62**, when assembling the cover **40** to the machine room **20**, the emboss **60** may be prevented from interfering with the lower end of the cover **40**.

[0156] The emboss **60** may be formed higher than the height of the plurality of main body hooks **30**. In other words, the height of the emboss **60** may be higher than the height of the main body hooks **30**. When the height of the emboss **60** is made higher than the height of the main body hooks **30**, the emboss **60** may be in close contact with the fixing portion **42** of the cover **40**. Accordingly, it is possible to prevent a gap from being formed between the support part **22** of the machine room

20 and the fixing portion **42** of the cover **40**.

[0157] Referring to FIG. **15**, the cover **40** may include a cover portion **41** and a fixing portion **42**.

[0158] The cover portion **41** may be formed in a roughly rectangular shape so as to cover the opening **21** of the machine room **20**. The cover portion **41** may include a plurality of air holes **45**. The plurality of air holes **45** may be formed as slits. External air may be introduced into the machine room **20** through the plurality of air holes **45**.

[0159] The fixing portion **42** may be provided at the edge of the cover portion **41**. The fixing portion **42** may be formed along the entire circumference of the cover portion **41**. The fixing portion **42** may be formed to correspond to the support part **22** of the machine room **20**.

[0160] In order to increase the rigidity of the cover **40**, the fixing portion **42** and the cover portion **41** may be formed to have a step. For example, the cover portion **41** may be formed as an approximately rectangular groove protruding from the fixing portion **42**.

[0161] The fixing portion **42** may include an upper fixing portion **421**, a lower fixing portion **422**, a left fixing portion **423**, and a right fixing portion **424**.

[0162] The upper fixing portion **421**, the left fixing portion **423**, and the right fixing portion **424** are the same as the above-described embodiment, so detailed descriptions thereof are omitted.

[0163] The lower fixing portion **422** may be provided on the lower portion of the cover **40**, that is, on the lower side of the cover portion **41**. The lower fixing portion **422** may be formed to correspond to the lower support part **222** of the machine room **20**. The lower fixing portion **422** may be in contact with the lower support part **222** of the machine room **20**.

[0164] The lower fixing portion **422** may include a plurality of cover catches **50**. In other words, the plurality of cover catches **50** may be provided at the lower portion of the cover **40**. The plurality of cover catches **50** may be provided below the cover portion **41**.

[0165] The plurality of cover catches **50** may be provided to correspond to the plurality of main body hooks **30** provided on one side of the main body **10**. In detail, the plurality of cover catches **50** may be formed so as to be caught on the plurality of main body hooks **30** provided on the lower support part **222** of the machine room **20**.

[0166] The plurality of cover catches **50** may be provided at regular intervals on the lower fixing portion **422** of the cover **40**. The plurality of cover catches **50** may be formed in the same shape.

[0167] Referring to FIG. **15**, the cover catch **50** may include a catch portion **54** and a guide portion **55**.

[0168] The catch portion **54** may protrude from one side of the cover **40** toward the machine room **20** of the main body **10**. The catch portion **54** may be formed approximately perpendicular to one side of the cover **40**. The catch portion **54** may be formed in an approximately rectangular shape. The catch portion **54** may be formed with the same thickness as the cover **40**.

[0169] The guide portion **55** may extend downward from the tip of catch portion **54**. The guide portion **55** may be formed so that the lower end of the guide portion **55** is positioned above the lower end of the cover **40**.

[0170] The guide portion **55** may be formed so that the width of the guide portion **55** becomes narrower from the upper end to the lower end. In other words, the width of the upper end of the guide portion **55** connected to the catch portion **54** may be larger than the width of the lower end of the guide portion **55**. For example, the guide portion **55** may be formed in an approximately triangular shape.

[0171] The guide portion **55** may be formed to be inclined downward from the tip of the catch portion **54**. When the guide portion **55** is formed to be inclined downward, the cover catch **50** may be easily connected with the main body hook **30**.

[0172] When the guide portion **55** of the cover catch **50** of the cover **40** is inserted into the catch groove **33** of the main body hook **30** of the main body **10**, the cover catch **50** and the main body hook **30** may be coupled. In this state, when the upper fixing portion **421** of the cover **40** is fixed to the upper support part **221** of the main body **10** using a fastening member, the cover **40** may be

assembled to the machine room **20** of the main body **10**.

[0173] Because there is the emboss **60** between the lower fixing portion **422** of the cover **40** and the lower support part **222** of the main body **10**, the lower fixing portion **422** of the cover **40** may be in close contact with the lower support part **222**. Accordingly, there may be no gap between the lower fixing portion **422** of the cover **40** and the lower support part **222** of the main body **10**. Therefore, when the compressor **11** operates, the vibration noise of the cover **40** may not occur.

[0174] As described above, the refrigerator **1** according to the disclosure may have the plurality of main body hooks **30** provided on one side of the main body **10** in which the machine room **20** is provided, the plurality of cover catches **50** provided on one side of the cover **40**, and the emboss **60** provided between the cover **40** and the main body **10**.

[0175] When the main body hooks **30** of the main body **10** and the cover catches **50** of the cover **40** are coupled, force may be applied to the cover **40** by the emboss **60** provided between the main body **10** and the cover **40**. Then, the cover **40** may be in close contact with the main body **10** so that there is no gap between the main body **10** and the cover **40**. In detail, when the emboss **60** is provided on the cover **40**, the tip of the emboss **60** is in close contact with the main body **10**, so that the vibration noise of the cover **40** may be prevented. When the emboss **60** is provided on the main body **10**, the tip of the emboss **60** is in close contact with the cover **40**, so that the vibration noise of the cover **40** may be prevented.

[0176] In the refrigerator **1** according to the disclosure having the structure as described above, the cover **40** is disposed in close contact with one side of the main body **10** in which the machine room **20** is disposed, so that the vibration noise of the cover **40** caused by the vibration of the compressor **11** may be prevented or minimized.

[0177] In addition, because the refrigerator **1** according to the disclosure may prevent the vibration noise by the cover **40** itself, the cost of the vibration-proof of the cover **40** may be reduced.

[0178] In the foregoing, the disclosure has been shown and described with reference to various embodiments. However, it is understood by those skilled in the art that various changes may be made in form and detail without departing from the scope of the disclosure as defined by the appended claims and equivalents thereof.

Claims

1. A refrigerator comprising: a machine room to be provided at a lower portion of a main body; an opening to be provided on one side of the main body to allow access to the machine room; a cover to be disposed on the one side of the main body to cover the opening; a plurality of cover catches to be provided on a lower portion of the cover; an emboss to be provided between the plurality of cover catches on the lower portion of the cover; and a plurality of main body hooks to be provided on the one side of the main body below the opening such that while the plurality of main body hooks are provided on the one side of the main body, locations of the plurality of main body hooks and the plurality of cover catches of the cover correspond to each other.

2. The refrigerator of claim 1, wherein while the opening of the machine room is covered with the cover, the plurality of cover catches are caught on the plurality of main body hooks, and the emboss contacts the one side of the main body on which the plurality of main body hooks are formed.

3. The refrigerator of claim 2, wherein a height of the emboss is greater than a gap between the cover and the one side of the main body.

4. The refrigerator of claim 1, wherein each of the plurality of main body hooks is formed to be inclined upwardly on the one side of the main body.

5. The refrigerator of claim 1, wherein the emboss includes a lower inclined surface formed to be inclined upwardly and facing a lower end of the cover.

6. The refrigerator of claim 5, wherein the emboss is formed to have an arc cross-section or a triangular cross-section.

7. The refrigerator of claim 1, wherein the emboss is among a plurality of embosses arranged in a straight line.

8. The refrigerator of claim 1, wherein the cover is formed as a rectangular flat plate, each of the plurality of cover catches is formed in a protrusion shape protruding from one side of the cover, and a catch groove into which a respective main body hook is insertable is provided at a lower end of a respective cover catch.

9. The refrigerator of claim 8, wherein the emboss is formed to protrude from one side of the cover along a direction that is same as a direction in which the plurality of cover catches protrude.

10. The refrigerator of claim 9, wherein a height of the emboss is higher than a height of at least one cover catch among the plurality of cover catches.

11. The refrigerator of claim 1, further comprising: a fastening member formed to fasten an upper portion of the cover to the one side of the main body above the opening.

12. The refrigerator of claim 1, wherein the main body comprises an upper support part, a lower support part, a left support part, and a right support part provided at an edge of the opening, the plurality of main body hooks are provided on the lower support part, and a plurality of screw holes are provided on the upper support part.

13. The refrigerator of claim 12, wherein the cover comprises an upper fixing portion and a lower fixing portion, the plurality of cover catches are provided on the lower fixing portion, and a plurality of bolt holes corresponding to the plurality of screw holes of the upper support part of the main body are provided on the upper fixing portion.

14. A refrigerator comprising: a machine room to be provided at a lower portion of a main body; an opening to be provided on one side of the main body to allow access to the machine room; a cover to be disposed on the one side of the main body to cover the opening; a plurality of main body hooks to be provided on the one side of the main body below the opening; an emboss to be provided on one side of the main body between the plurality of main body hooks and below the opening; and a plurality of cover catches to be provided on a lower portion of the cover such that while the cover is disposed on the one side of the main body, locations of the plurality of cover catches and the plurality of main body hooks of the main body correspond to each other.

15. The refrigerator of claim 14, wherein each of the plurality of cover catches is formed to be inclined downward on a lower portion of the cover, each of the plurality of main body hooks is formed in a protrusion shape protruding from one side of the main body, and a catch groove into which the cover catch is inserted is provided on an upper surface of the main body hook.
